

APPLICATIONS:

A muffle furnace is a high-temperature, front-loading oven used for controlled thermal processing. It isolates the material being heated from direct combustion gases or external contaminants. Its defining advantage is providing uniform, contamination-free, and precise heating.

HEATING ELEMENT

Muffle furnace fabrication involves constructing an external metal casing (typically heavy-duty mild steel or stainless steel), layering the interior with high-grade ceramic fiber or refractory brick for thermal insulation, and installing heating elements like Kanthal A1 & Silicon Carbide Rod (up to 1200°C / 1400°C).



FABRICATION

Fabricated on heavy M.S plates with heavy double walled door with ceramic wool used as insulation. Inner made of Ceramic fiber and Brick insulation, External made of CRCA/MS duly epoxy powder coated. Door insulation of HFK bricks.

HEATING ELEMENT

Imported KANTHAL A1 & SILICON CARBIDE ROD Heating elements are provided all around the heating chamber backed by CERAMIC FIBER or Graded Insulation Brick Work. A Safety protection fuse is provided to prevent the heating elements from accidental heating. A simple flap type of door with a spy hole for viewing the chamber.

TEMPERATURE CONTROL

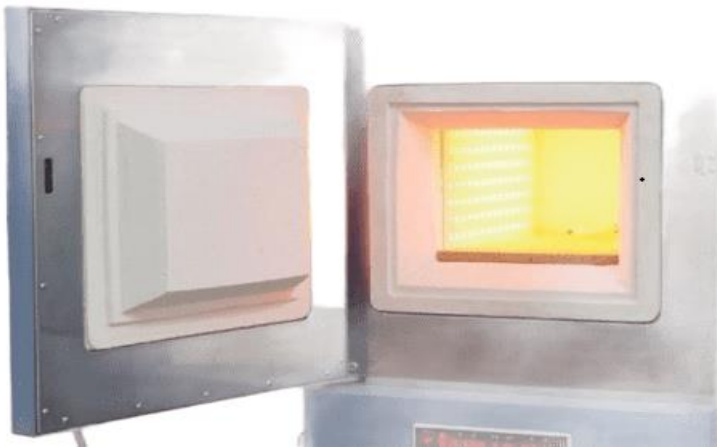
- *Microprocessor based PID temperature controller with dual display (Process value and set value), low and high limit setting with Thermocouple CRAL 'K' type timer control.*

POWER SUPPLY

230 Volts Single Phase 50Hz, AC supply.

FEATURES

- *Sturdy Construction*
- *High operational fluency*
- *Low energy consumption*
- *Impeccable performance*
- *Digital Temperature Controller*
- *Double Walled Design*



TEMPERATURE RANGE

Muffle furnaces are available with a range of maximum temperatures.

The maximum temperature is dependent on the type of heating elements used in the construction.

The most common have metallic wire heating elements with a maximum temperature in the range of 900°C to 1200°C / 1400°C.

CERTIFICATIONS

- *CE Certified*
- *Quality Certificate: ISO 9001:2015*

Model	Capacity	Inner Chamber Size (WxDxH)	Load kW
RS-MF-2	2 Liters	4 x 4 x 9 inch	1.5
RS-MF- 7	7 Liters	8 x 9 x 6 inch	3
RS-MF- 12	12 Liters	9.5 X 12.2 X 6.3 inch	3.5
RS-MF-17	17 Liters	9.8 x 15.8 x 6.7 Inch	4
RS-MF-20	20 Liters	11.2 x 15.75 x 7.09	4.5

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